Lecture 20

Fundamentals of Numerical Analysis

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Recap: Integrals

- Often $f(x)$ is only known implicitly (known at a certain number of points); often the anti-derivative of $f(x)$ is not known.
- The Fundamental Theorem of Calculus states that $\int_a^b f(x)\,dx = F(b) - F(a)$, where F is the antiderivative of f. We don't know F, so we approximate the integral operation.
- Lower Sum and Upper Sum:

$$L(f;P) = \sum_{i=0}^{n-1} m_i(x_{i+1} - x_i), \qquad U(f;P) = \sum_{i=0}^{n-1} M_i(x_{i+1} - x_i)$$

$$L(f;P) \leqslant \int_a^b f(x)\,dx \leqslant U(f;P)$$

In the limit, they are equal

$$\lim_{n\to\infty} L(f;P) = \int_a^b f(x)\,dx = \lim_{n\to\infty} U(f;P)$$

- Generally the Riemann sum is

$$S = \sum_{i=0}^{n-1} f(z_i)(x_{i+1} - x_i), \qquad x_i \leqslant z_i \leqslant x_{i+1}$$

$z_i = x_i$ is a Left Riemann Sum, $z_i = x_{i+1}$ is a Right Riemann Sum, $z_i = \frac{x_{i+1}+x_i}{2}$ is a Middle Riemann Sum

Recap: Trapezoid

- Old Idea: Left,Right,Midpoint Riemann integration approximate $f(x)$ by a constant function and obtain the area under the constant function.
- New Idea: Trapezoid approximates $f(x)$ by a linear function (degree one polynomial) and obtains the area under the linear function.

Basic Trapezoid

$$\int_a^b f(x)\,dx \approx \frac{1}{2}(b-a)(f(a)+f(b))$$

Composite Trapezoid

With uniform spacing of P, $h_i = x_{i+1} - x_i = h$ is constant

$$T(f;P) = \int_a^b f(x)\,dx \approx \frac{h}{2}\left(f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)\right)$$

Recap: Trapezoid Accuracy

- Order of accuracy: look at h^p with $p \approx \frac{log(err^{(k)}/err^{(k-1)})}{log(h^{(k)}/h^{(k-1)})}$
- Basic Trapezoid error: $E = -\frac{(b-a)^3 f''(\eta)}{12}$
- Each subinterval $[x_i, x_{i+1}]$ has local error $E_i = -\frac{h^3}{12} f''(\eta_i)$, so summing over all subintervals:

$$\boxed{E = -\frac{(b-a)h^2}{12} f''(\xi)}$$

Hence, the composite trapezoidal rule is second–order accurate: $E = O(h^2)$.

How do we improve Trapezoid?

- instead of a linear approximation, use a quadratic approximation
- $\Rightarrow$ Simpson's Rule

Simpson

- consider $\int_a^b f(x)\,dx$
- partition $P = \{a, a+h, b = a+2h\}$
- or $\int_0^{2h} f(x)\,dx$
- partition $P = \{0, h, 2h\}$
- replace $f(x)$ by a quadratic $p(x)$:

$$\begin{aligned} f(x) &\approx p(x) \\ &= f(0) + \frac{f(h) - f(0)}{2}x + \frac{f(2h) - 2f(h) - f(0)}{2h^2}x(x-h) \\ &= \text{newton form} \end{aligned}$$

- integrate $\int_0^{2h} p(x)\,dx$:

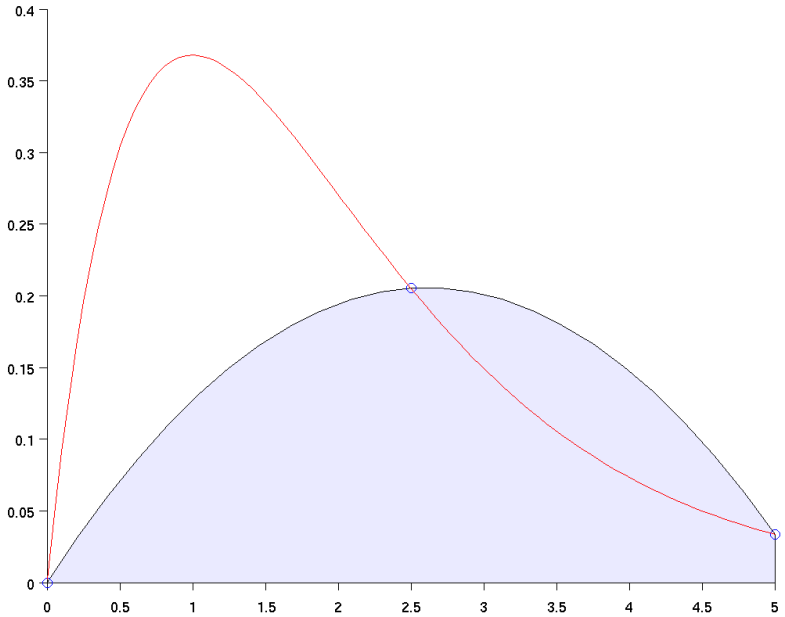
$$\begin{aligned} \int_0^{2h} f(x)\,dx &\approx \int_0^{2h} p(x)\,dx \\ &= \frac{h}{3}\left[f(0) + 4f(h) + f(2h)\right] \end{aligned}$$

Simpson

Since $b - a = 2h$ we have

basic Simpson's Rule

$$\int_a^b f(x)\,dx \approx \frac{b-a}{6}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right]$$



Composite Simpson

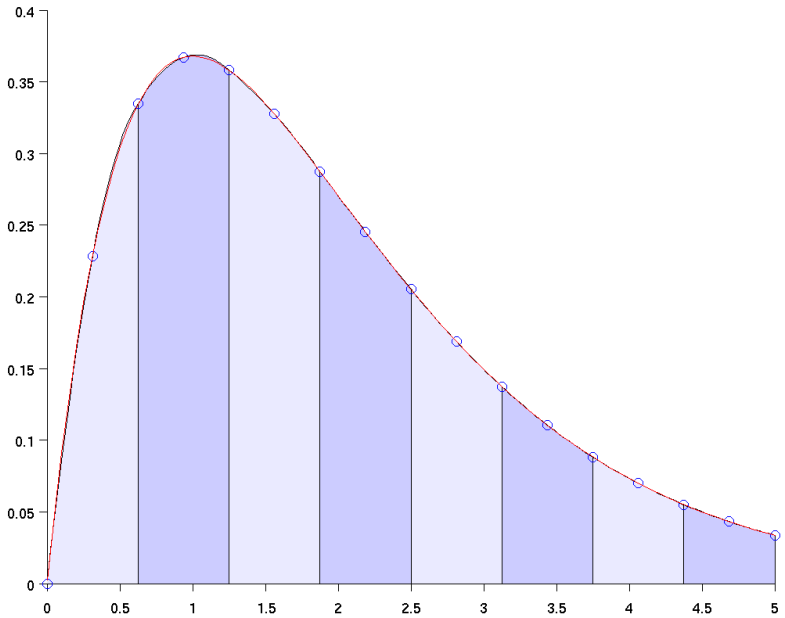
Over a uniform partition $P = x_0, x_1, \ldots, x_n$, use Basic Simpson's Rule over each subinterval $[x_{2i}, x_{2i+2}]$

$$\begin{aligned}
\int_a^b f(x)\,dx &= \sum_{i=0}^{n/2-1} \int_{x_{2i}}^{x_{2i+2}} f(x)\,dx \\
&= \sum_{i=0}^{n/2-1} \frac{2h}{6}\left[f(x_{2i}) + 4f(x_{2i+1}) + f(x_{2i+2})\right] \\
&= \frac{h}{3}\left[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + \cdots + 4f(x_{n-1}) + f(x_n)\right]
\end{aligned}$$

Simpson

Composite Simpson's Rule

$$\int_a^b f(x)\,dx \approx \frac{h}{3}\left[f(a)+f(b)+4\sum_{i=1}^{n/2} f(a+(2i-1)h)+2\sum_{i=1}^{n/2-1} f(a+2ih)\right]$$



Simpson

How accurate is Simpson? Recall composite Trapezoid:

$$I - T(f, P) = -\frac{1}{12}(b-a)h^2 f''(\xi) - \mathcal{O}(h^2)$$

Prediction? $\mathcal{O}(h^2)$? $\mathcal{O}(h^3)$? $\mathcal{O}(h^4)$?

`int_simpson_test.m`

Why is composite Simpson $\mathcal{O}(h^4)$?

Taylor Series:

$$f(a+h) = f + hf' + \frac{1}{2!}h^2f'' + \frac{1}{3!}h^3f''' + \frac{1}{4!}h^4f^{(4)} + \frac{1}{5!}h^5f^{(5)} + \dots$$

$$f(a+2h) = f + 2hf' + 2h^2f'' + \frac{4}{3}h^3f''' + \frac{2}{3}h^4f^{(4)} + \frac{4}{15}h^5f^{(5)} + \dots$$

This gives

$$\frac{h}{3}\left[f(a) + 4f(a+h) + f(b)\right] = 2hf + 2h^2f' + \frac{4}{3}h^3f'' + \frac{2}{3}h^4f''' + \frac{5}{18}h^5f^{(4)}$$

Integrating the Taylor Series expansion of $f(x)$ exactly gives

$$\int_a^b f(x)\,dx = 2hf + 2h^2f' + \frac{4}{3}h^3f'' + \frac{2}{3}h^4f''' + \frac{4}{15}h^5f^{(4)}$$

So basic Simpson's Rule gives an error of

$$-\frac{1}{90}\left(\frac{b-a}{2}\right)^5 f^{(4)}(\xi)$$

Why is composite Simpson $\mathcal{O}(h^4)$?

basic Simpson's Rule:

$$-\frac{1}{90}\left(\frac{b-a}{2}\right)^5 f^{(4)}(\xi)$$

Over $n/2$ subintervals $[x_{2i}, x_{2i+2}]$ becomes:

$$\begin{aligned} err &= \sum_{i=1}^{n/2} -\frac{1}{90}\left(\frac{x_{2i+2}-x_{2i}}{2}\right)^5 f^{(4)}(\xi_i) = -\frac{1}{90}\sum_{i=1}^{n/2}\left(\frac{2h}{2}\right)^5 f^{(4)}(\xi_i) \\ &= -\frac{1}{90}\frac{n}{2}h^5 f^{(4)}(\xi) = -\frac{1}{180}\frac{(b-a)}{h}h^5 f^{(4)}(\xi) \\ &= -\frac{b-a}{180}h^4 f^{(4)}(\xi) \end{aligned}$$

Composite Simpson's Rule

$$-\frac{b-a}{180}h^4 f^{(4)}(\xi)$$

We "gain" two orders over Trapezoid

Can we generalize?

Summary:

- left/right Riemann: approximate $f(x)$ by 0-degree $p(x)$ and integrate
- Trapezoid: approximate $f(x)$ by 1-degree $p(x)$ and integrate
- Simpson: approximate $f(x)$ by 2-degree $p(x)$ and integrate

Degree of Precision

If the integration rule has zero error when integrating any polynomial of degree $\leqslant r$

and

If the error is nonzero for some polynomial of degree $r+1$,
and

then, the rule has *degree of precision* equal to r.

(basic) Newton-Cotes rules:

name	n	formula
Trapezoid	1	$\frac{(b-a)}{2}[f(a)+f(b)]$
Simpson's $1/3$	2	$\frac{(b-a)}{6}\left[f(a)+4f(\frac{a+b}{2})+f(b)\right]$
Simpson's $3/8$	3	$\frac{(b-a)}{8}[f(a)+3f(a+h)+3f(b-h)+f(b)]$
Boole's	4	$\frac{(b-a)}{90}\left[7f(a)+32f(a+h)+12f(\frac{a+b}{2})+32f(b-h)+7f(b)\right]$

Error bounds for composite Newton-Cotes:

name	n	error	h
Trapezoid	1	$-\frac{(b-a)h^2}{12}f''(\xi)$	$h=b-a$
Simpson's $1/3$	2	$-\frac{(b-a)h^4}{90}f^{(4)}(\xi)$	$h=(b-a)/2$
Simpson's $3/8$	3	$-\frac{(b-a)h^4}{80}f^{(4)}(\xi)$	$h=(b-a)/3$
Boole's	4	$-\frac{2(b-a)h^6}{945}f^{(6)}(\xi)$	$h=(b-a)/4$

Quadrature

- up until now, our quadrature methods were of the form

$$\int_a^b f(x)\,dx \approx \sum_{j=0}^{n} w_j f(x_j)$$

  where x_j are equally spaced nodes
- Trapezoid:

$$\int_a^b f(x)\,dx \approx \frac{b-a}{2} f(a) + \frac{b-a}{2} f(b)$$

- Simpson:

$$\int_a^b f(x)\,dx \approx \frac{b-a}{6} f(a) + \frac{2(b-a)}{3} f\left(\frac{a+b}{2}\right) + \frac{b-a}{6} f(b)$$

- Similar for higher order polynomial Newton-Cotes rules

Quadrature with Freedom

!

These quadrature rules have one thing in common: they're restrictive

- e.g. Simpson:

$$\int_a^b f(x)\,dx \approx \frac{b-a}{6} f(a) + \frac{2(b-a)}{3} f\left(\frac{a+b}{2}\right) + \frac{b-a}{6} f(b)$$

- Trapezoid, Simpson, etc (Newton-Cotes) are based on equally spaced nodes

- We know one thing already from interpolation: equally spaced nodes result in *wiggle*.
- What other choice do we have? (...recall how we fixed wiggle in interpolation: by moving the location of the nodes)

Gaussian Quadrature

- free ourselves from equally spaced nodes
- combine selection of the nodes and selection of the weights into one quadrature rule

Gaussian Quadrature

Choose the nodes and coefficients optimally to maximize the degree of precision of the quadrature rule:

$$\int_a^b f(x)\,dx \approx \sum_{j=0}^{n} w_j f(x_j)$$

Goal

Seek w_j and x_j so that the quadrature rule is exact for really high polynomials

Gaussian Quadrature

$$\int_a^b f(x)\,dx \approx \sum_{j=0}^{n} w_j f(x_j)$$

- we have $n+1$ points $x_j \in [a,b]$, $a \leqslant x_0 < x_1 < \cdots < x_{n-1} < x_n \leqslant b$.
- we have $n+1$ real coefficients w_j

- so there are $2n+2$ total unknowns to take care of

- there were only 2 unknowns in the case of trapezoid (2 weights)
- there were only 3 unknowns in the case of Simpson (3 weights)
- there were only $n+1$ unknowns in the case of general Newton-Cotes ($n+1$ weights)

Gaussian Quadrature

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- there were only 3 unknowns in the case of Simpson (3 weights)
- there were only $n+1$ unknowns in the case of general Newton-Cotes ($n+1$ weights)

$2n+2$ unknowns (using $n+1$ nodes) can be used to exactly interpolate and integrate polynomials of degree up to $2n+1$

Better Nodes Example

The first thing we do is SIMPLIFY

- consider the case of $n = 1$ (2-point)
- consider $[a, b] = [-1, 1]$ for simplicity
- we *know* how the trapezoid rule works
- Question: can we possibly do better using only 2 function evaluations?
- Goal: Find w_0, w_1, x_0, x_1 so that

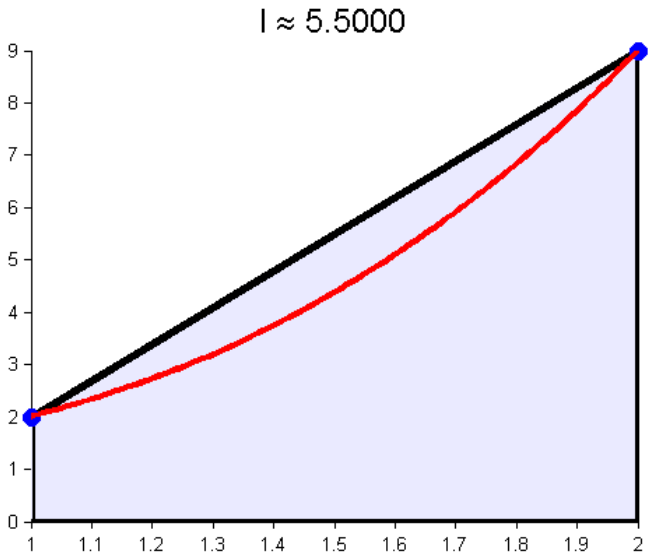
$$\int_{-1}^{1} f(x)\,dx \approx w_0 f(x_0) + w_1 f(x_1)$$

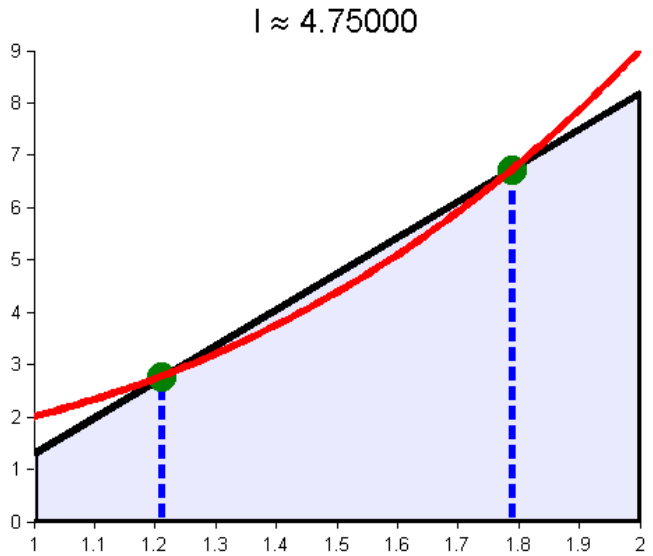
is as accurate as possible...

Graphical View

Consider

$$\int_1^2 x^3 + 1\, dx = 4.75$$





Derive...

Again, we are considering $[a, b] = [-1, 1]$ for simplicity:

$$\int_{-1}^{1} f(x)\, dx \approx w_0 f(x_0) + w_1 f(x_1)$$

Goal: find w_0, w_1, x_0, x_1 so that the approximation is exact up to cubics. So try any cubic:

$$f(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3$$

This implies that:

$$\begin{aligned} \int_{-1}^{1} f(x)\, dx &= \int_{-1}^{1} \left(c_0 + c_1 x + c_2 x^2 + c_3 x^3\right)\, dx \\ &= w_0 \left(c_0 + c_1 x_1 + c_2 x_1^2 + c_3 x_1^3\right) + \\ &\quad w_1 \left(c_0 + c_1 x_2 + c_2 x_2^2 + c_3 x_2^3\right) \end{aligned}$$

Derive...

$$\begin{aligned}\int_{-1}^{1} f(x)\,dx &= \int_{-1}^{1} \left(c_0 + c_1 x + c_2 x^2 + c_3 x^3\right)\,dx \\ &= w_0\left(c_0 + c_1 x_1 + c_2 x_1^2 + c_3 x_1^3\right) + \\ &\quad w_1\left(c_0 + c_1 x_2 + c_2 x_2^2 + c_3 x_2^3\right)\end{aligned}$$

Rearrange into constant, linear, quadratic, and cubic terms:

$$\begin{aligned}c_0\left(w_0 + w_1 - \int_{-1}^{1} dx\right) &+ c_1\left(w_0 x_0 + w_1 x_1 - \int_{-1}^{1} x\,dx\right) + \\ c_2\left(w_0 x_0^2 + w_1 x_1^2 - \int_{-1}^{1} x^2\,dx\right) &+ c_3\left(w_0 x_0^3 + w_1 x_1^3 - \int_{-1}^{1} x^3\,dx\right) = 0\end{aligned}$$

Since c_0, c_1, c_2 and c_3 are arbitrary, then their coefficients must all be zero.

Derive...

This implies:

$$w_0 + w_1 = \int_{-1}^{1} dx = 2 \qquad w_0 x_0 + w_1 x_1 = \int_{-1}^{1} x\, dx = 0$$

$$w_0 x_0^2 + w_1 x_1^2 = \int_{-1}^{1} x^2\, dx = \frac{2}{3} \qquad w_0 x_0^3 + w_1 x_1^3 = \int_{-1}^{1} x^3\, dx = 0$$

Some algebra leads to:

$$w_0 = 1 \quad w_1 = 1 \quad x_0 = -\frac{\sqrt{3}}{3} \quad x_1 = \frac{\sqrt{3}}{3}$$

Therefore:

$$\int_{-1}^{1} f(x)\, dx \approx f\left(-\frac{\sqrt{3}}{3}\right) + f\left(\frac{\sqrt{3}}{3}\right)$$